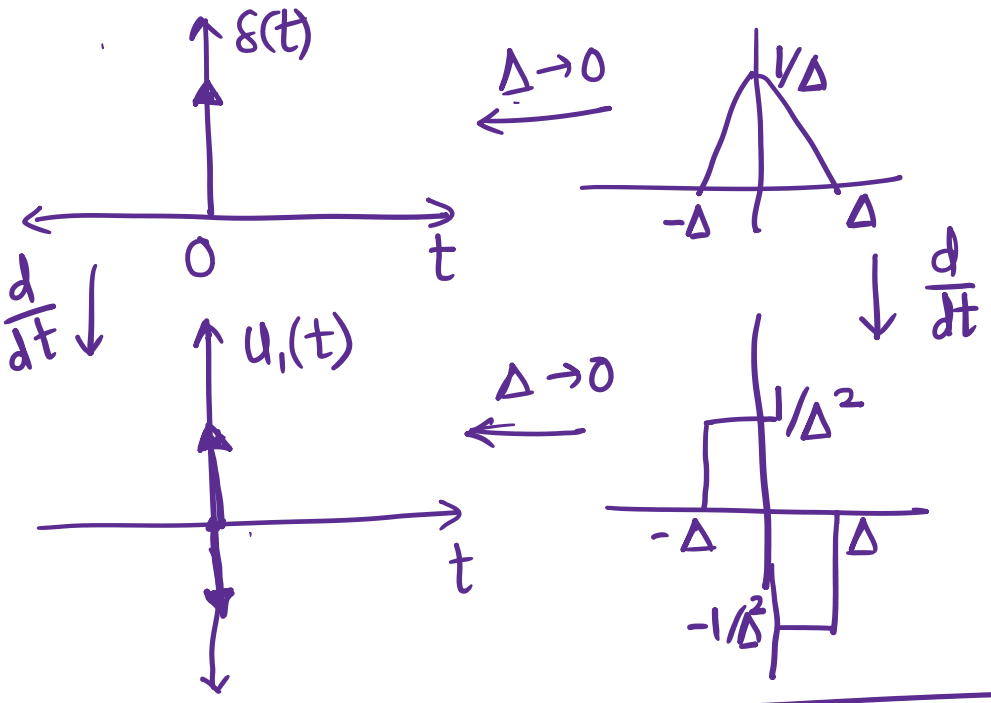
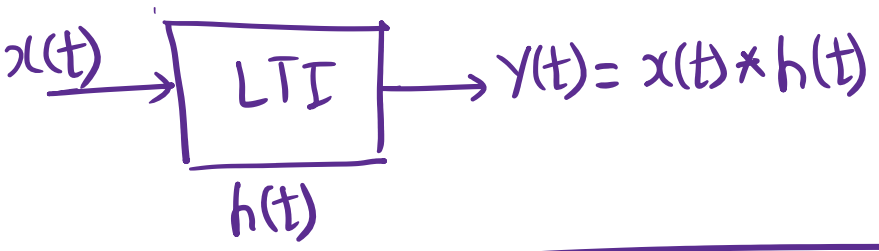
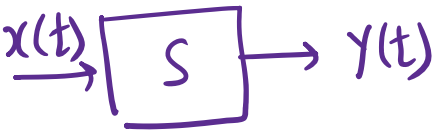




Differential Eq^n : (LCCDE)

$$\sum_{k=0}^{M} a_k \frac{d^k y}{dt^k} = \sum_{k=0}^{N} b_k \frac{d^k x}{dt^k} \quad , \text{ order} = \max(M, N)$$

$x(t)$: i/p & $y(t)$: o/p



$$\frac{d^2 y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + x(t) \quad , x(t) = e^{2t} u(t)$$

$y(0) = 0$

$y'(0) = 0$

$$y(t) = y_h(t) + y_p(t)$$

$$x(t) = e^{st} \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = ? = x(t) * h(t)$$

$h(t)$

$$= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau$$

$$= e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau}$$

$$y(t) = e^{st} \quad H(s)$$

$$e^{st} \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = H(s)\, e^{st}$$

$h(t)$ eigenvalue eigenfunct^n

$$Ax = \lambda x$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{s_k t} \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = \sum_{k=-\infty}^{\infty} a_k H(s_k)\, e^{s_k t}$$

$h(t)$

Discrete Time Fourier Series (DTFS)

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} \longrightarrow$$

$x[n]$: Discrete Time N-periodic finite power signal

$$S_N \triangleq \left\{ x : \mathbb{Z} \longrightarrow \mathbb{C} : x[n+N] = x[n]\ \forall n \ \&\ \frac{1}{N} \sum_{\langle N \rangle} |x[n]|^2 < \infty \right\}$$

To show: S_N is a V.S. & inner product space with $\langle x, y \rangle \triangleq \frac{1}{N} \sum_{\langle N \rangle} x[n]\, y^*[n]$

Proof: $(x+y)[n] \triangleq x[n]+y[n]$

$(\alpha x)[n] \triangleq \alpha x[n]$

i) If $x, y \in S_N \Rightarrow \underbrace{x+y}\in S_N$ (To show)

✓ $(x+y)[n+N] = x[n+N]+y[n+N] = x[n]+y[n] = (x+y)[n]$

ii) $$\frac{1}{N}\sum_{\langle N\rangle}|(x+y)[n]|^2 = \frac{1}{N}\sum_{\langle N\rangle}|x[n]+y[n]|^2$$
$$\leq \frac{1}{N}\sum_{\langle N\rangle}(|x[n]|+|y[n]|)^2$$
$$\leq \frac{1}{N}\Big(\underbrace{\sum_{\langle N\rangle}|x[n]|^2+\sum_{\langle N\rangle}|y[n]|^2}_{<\infty}+\underbrace{\sum_{\langle N\rangle}2|x[n]||y[n]|}_{<\infty}$$
$$\leq \infty$$

$\therefore\ x+y \in S_N$

II $(\alpha x)[n+N] = \alpha x[n+N] = \alpha x[n] = (\alpha x)[n]$

$\frac{1}{N}\sum_{\langle N\rangle}|(\alpha x)[n]|^2 = \frac{|\alpha|^2}{N}\sum_{\langle N\rangle}|x[n]|^2 < \infty$ $\Big\}$ $\alpha x \in S_N$

$\Rightarrow$ S_N is a V.S.!

$\langle x,y\rangle = \frac{1}{N}\sum_{\langle N\rangle}x[n]y^*[n]$. $\rightarrow$ Inner Product

---

$\phi : S_N \longrightarrow \mathbb{C}^N \longrightarrow$ N dimens^n

$x \longrightarrow \phi(x) \triangleq \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix}$ $\Big\}$ Dimens^n of $S_N = N$

$x_k[n] \triangleq e^{jk\frac{2\pi}{N}n}$ $\qquad \frac{1}{N}\sum_{\langle N\rangle}|x_k[n]|^2 = 1$

$x_k \in S_N$

$x_k \in S_N$

$\{x_k\}_{k=0}^{N-1}$,

$\Downarrow$

Set of basis elements

$$\Rightarrow x = \sum_{k=0}^{N-1} a_k x_k$$

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\frac{2\pi}{N}n}$$

$$\langle x_k, x_l \rangle = \frac{1}{N} \sum_{\langle N \rangle} x_k[n] x_l^*[n]$$

$$= \frac{1}{N} \sum_{\langle N \rangle} e^{jk\frac{2\pi}{N}n} \cdot e^{-jl\frac{2\pi}{N}n}$$

$$= \frac{1}{N} \sum_{\langle N \rangle} e^{j(k-l)\frac{2\pi}{N}n}$$

$$= \begin{cases} 1, & k = l \\ 0, & k \neq l \end{cases} \text{ (Roots of unity)}$$